

Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title:

SmartResume Generator

Team Name:

GENZAI

Team Members:

- Kondaveeti Lakshmi Lavanya
 - Bejjam Manasa
 - Pedapudi Swathi
 - Kanamarlapudi Sree Satya Sai Chandrika
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Phase-1: Brainstorming & Ideation

Objective:

Smart Resume Generator is to streamline and enhance the resume creation process using automation and intelligent features.

Key Points:

- 1) **Problem Statement:**
 - In today's competitive job market, job seekers often struggle with creating a professional, well-structured, and ATS-friendly resume.
 - A Smart Resume Generator will leverage automation and AI to help users quickly create customized, professional, and ATS-optimized resumes

2) **Proposed Solution:**

- Automated Content Suggestions – Uses AI to generate job descriptions, skills, and achievements based on the user's job role and experience.
- Grammar & Readability Enhancement – Provides suggestions to improve clarity, tone, and professionalism.

3) **Target Users:**

- Fresh Graduates & Students – Need guidance in creating their first professional resume with limited work experience.
- Entry-Level Professionals – Require well-structured resumes to stand out in a competitive job market.

4) **Expected Outcome:**

- Users can generate a professional-quality resume in minutes instead of hours.

Phase-2: Requirement Analysis

Objective:

- Define technical and functional requirements.

Key Points:

1. **Technical Requirements:**

- Programming Language: **python**
- Frontend: **Frameworks using React.js, Next.js**
- Backend: **Authentication by JWT-based authentication**
- Database: **PostgreSQL, MongoDB**

2. **Functional Requirements:**

- Users should be able to sign up for an account using email, Google, or LinkedIn. Account should store user profile and resume data.
Authentication & Authorization
- Users should be able to choose from multiple pre-designed templates based on job roles, industries, or personal preferences.
- Users should be able to see a live preview of their resume as they edit, making it easier to visualize the changes instantly.

3. **Constraints & Challenges:**

- **Technical Constraints:** AI-generated resume content (e.g., job descriptions, skills) may not always be accurate, industry-specific, or contextually relevant.
- **Real-Time Processing & Performance:** Generating real-time previews and AI-powered suggestions while ensuring a smooth user experience can be resource-intensive.

Phase-3: Project Design

Objective:

Create the architecture and user flow.

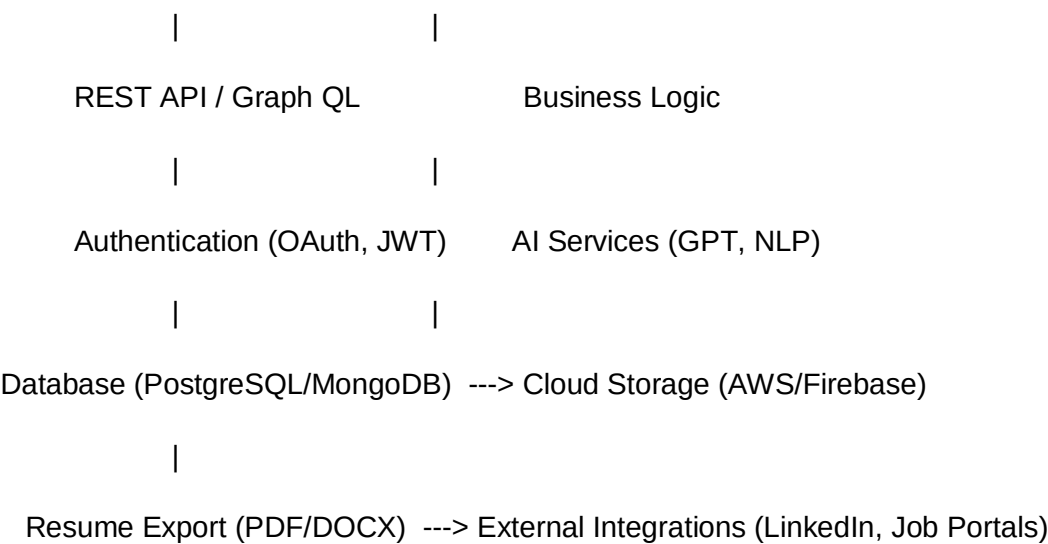
Architecture Diagram (High-Level Components)

pgsql

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User (Web/App) ---> Frontend (React.js/Next.js) ---> Backend (Node.js/FastAPI)



Key Points:

1. **System Architecture Diagram:**
 - User authentication (Google OAuth, Email, etc)Resume template selection Form-based resume builder
 - User authentication & session management AI-powered text generation (OpenAI API, LLM models)Resume parsing & formatting

2. User Flow:

- **Step 1:** Choose a Resume Template User is presented with multiple professional templates. Option to select a template based on industry or job type.
- **Step 2:** Enter Personal Information Sections include: Full Name, Contact Info Job Title & Professional Summary Work Experience (Job Titles, Companies, Dates) Education Details (Degree, Institution, Year) Skills
- **Step 3:** AI Optimization & Customization AI analyzes input and suggests Stronge wording for bullet points.
- **Step 4:** Real-Time Resume Preview Users can see a live preview of their resume.

3. UI/UX Considerations:

- **Onboarding & Navigation:** Simple & Intuitive Navigation Clear call-to-action (CTA) buttons like “Create Resume”, “Upload Resume”, and “Download”.
- **Resume Creation Interface:** Minimalist & Clean UI Form-based layout with clear field labels and input examples. Auto-save feature to prevent data loss.
- **Resume Preview & Customization:** Live Resume Preview Side-by-side preview that updates in real time.

Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down the tasks using Agile methodologies.

Sprint	Task	Priority	Duration	Deadline	Assigned to	Expected outcomes
Sprint 1	python	High	6 hours (day1)	End of Day 1	Kondaveeti Lakshmi Lavanya	Working of code by python
Sprint 2	Frontend: Frameworks	Medium	2 hours (Day1)	End of Day 1	Bejjam Manasa	Frameworks by reac.js
Sprint 3	Backend: Authentication by JWT-based authentication	Medium	3 hours (Day2)	Mid-day 2	Pedapudi Swathi	Authentication by JWT-based authentication
Sprint 4	Database: PostgreSQL, MongoDB	Medium	2 hours (Day2)	End of Day 2	Kanamarlapudi Sree Satya Sai Chandrika	Using MongoDB

Phase-5: Project Development

Objective:

- Code the project and integrate components.

Key Points:

Technology Stack Used:

- Programming Language: **python**
- Frontend: **Frameworks using React.js, Next.js**
- Backend: **Authentication by JWT-based authentication**
- Database: **PostgreSQL, MongoDB**

1. Development Process:

- Step 1: User logs in/signs up.
- Step 2: Selects a template or starts from scratch.
- Step 3: Enters personal information, education, experience, and skills.
- Step 4: AI suggests improvements.
- Step 5: User customizes design elements.
- Step 6: Preview & Download.

2. Challenges & Fixes:

- **Technical Constraints:** AI-generated resume content (e.g., job descriptions, skills) may not always be accurate, industry-specific, or contextually relevant.
- **Real-Time Processing & Performance:** Generating real-time previews and AI-powered suggestions while ensuring a smooth user experience can be resource-intensive

Phase-6: Functional & Performance Testing

Objective:

- Ensure the project works as expected.

Test Case ID	Category	Test Scenario	Expected Outcomes	Status	Tester
TC-001	Functional testing	Resume has been created	Frame works are created	passed	Tester 1
TC-002	Functional testing	Resume has in correct manner	Frameworks, data are in correct manner	passed	Tester 2
TC-003	Representation of task	Resume has in the correct information	Data had given with correct manner	Need optimization	Tester 3
TC-004	Final validation	Resume has been approved	Data is in correct manner	fixed	Tester 2

Final Submission

1. **Project Report Based on the templates**
 2. **Demo Video (3-5 Minutes)**
 3. **GitHub/Code Repository Link**
 4. **Presentation**
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